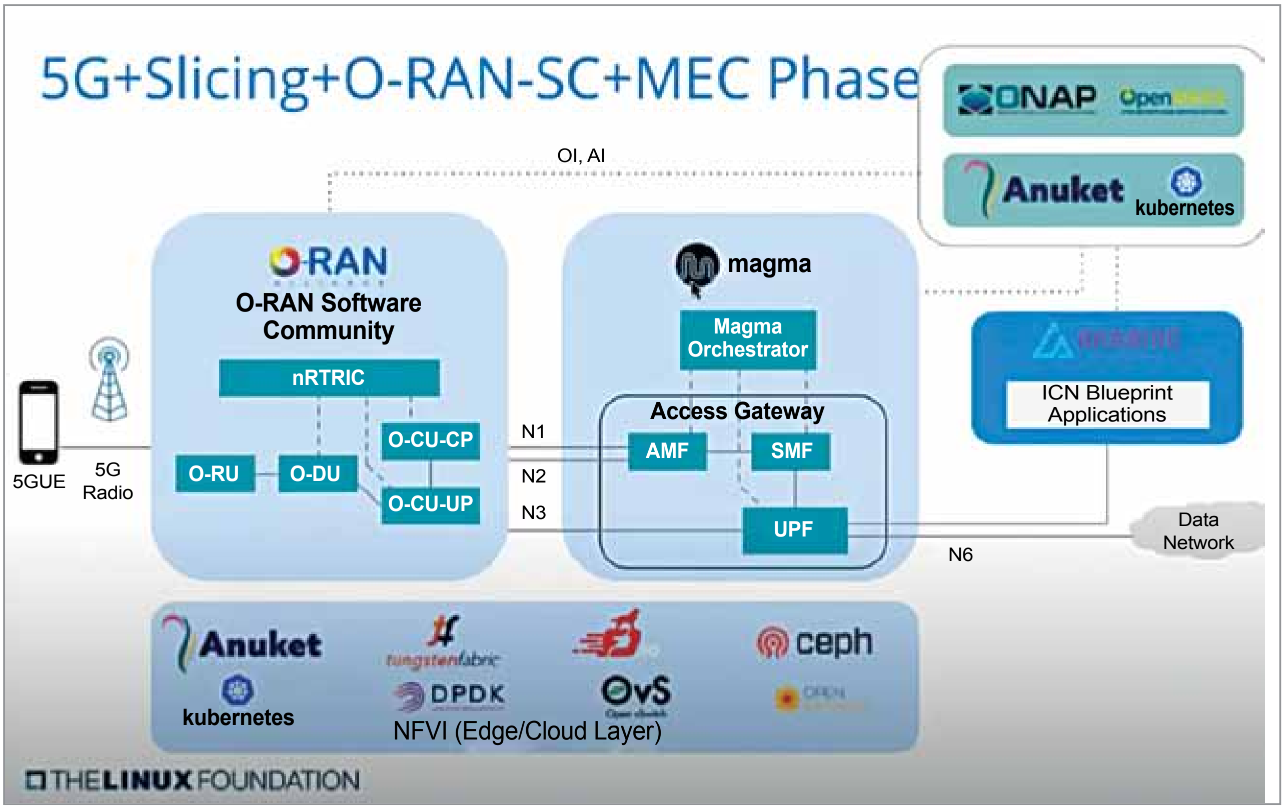

Figure 4: The third phase of 5G modernisation

the community or their vendors. So, instead of multiple vendors doing things in different ways, now you have it done in the community and then the vendor can provide innovation and differentiation on top.

5G will modernise today's energy grid with awareness monitoring across distribution systems.

It will roll out in three phases: 5G core+MEC phase, 5G E2E+MEC phase, and 5G+Slicing+O-RAN-SC +MEC phase.

In the first phase the communities are using the emulators on the RAN side, but Magma is being inserted with ONAP and Anuket on Kubernetes. There are some components from Intel called openness also coming in but, in general, those are the big new components.

The second phase is bringing in a commercial genome piece. The RAN side of things is integrated, and then you run a slice through it. So, the end-to-end network slicing comes in.

The last phase, which is towards the end of the year, is when ORAN comes in. What's happened is that ONAP and ORAN are already deeply integrated with interfaces. There is a lot of component reuse between projects because these are all microservices, containerised modules, and you can just use them as needed and it will just get you a head start. And that's already happening.

Summing up

Completing the solution stacks to integrate a fully open networking and edge stack, speeding up 'backend interior and deployment process' with open compliance, integrated deployment in vertical markets to start embracing, utilising and building on networking, cloud, edge stacks are seen as the top priorities.

To conclude, blueprints are fairly matured and deployed. Most of the projects are live in production. The first step is that it needs to happen. In several use cases, the 5G node is now merged from the open-ended interface side of things. And so, Magma itself, as an open mobile packet core in the Q3 timeframe, will be fairly mature.

The testing of these blueprints should be ready by the end of the year. Till then it is a proprietary mobile packet core. Pretty much everything that is at the higher layers of the stack is mature. At the lower layers, radio will stay proprietary, but the disaggregation concept is mature. END

This article is based on the tech talk by Arpit Joshipura, GM, networking edge and IoT, Linux Foundation, at Open Source Conference 2021 (Virtual).

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